

Abhay Yadav

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SUMMARY

AI engineer building LLM systems with measurable accuracy rather than demo-grade behaviour — agentic workflows, retrieval over domain documents, and natural-language-to-query generation. Designs the evaluation harness before the agent, keeps deterministic logic out of the model, and reports precision separately from coverage. Comfortable owning a system end to end: retrieval, tool use, guardrails, and the benchmark that proves it works.

EDUCATION

G.L. Bajaj Institute of Technology and Management <i>B.Tech in Computer Science and Engineering</i>	2023 – 2027 <i>CGPA: 8.1 / 10</i>
L.B.P. School, CBSE (Kota) <i>Class XII</i>	2022 – 2023 <i>75%</i>
Ambuja Public School, CBSE (Pali) <i>Class X</i>	2020 – 2021 <i>92%</i>

TECHNICAL SKILLS

Languages: Python, SQL, SOQL, JavaScript
AI/ML: LLM Agents, RAG, MCP, Tool Calling, Prompt Engineering, Embeddings, Evaluation & Benchmarking
Backend: FastAPI, PostgreSQL, pgvector, REST APIs
Tools: Docker, Git, GitHub Actions, pytest, React

PROJECTS

- AI Finance Controller** — Agentic Invoice Reconciliation | *Python, FastAPI, PostgreSQL, Claude API, pytest*
- Built an agent that reconciles settlement lines against ledger entries, generating planted ground truth **before** the agent existed so accuracy could be measured rather than asserted.
 - Kept the model out of arithmetic by design — it selects candidates but never computes amounts; a `--no-llm` mode runs the full deterministic pipeline as a regression baseline.
 - Derived the auto-post confidence threshold from a calibration table instead of picking one, and reported **match precision separately from match rate** (98.3% match rate at 100% precision).
 - Enforced module layering in CI so matching cannot import adjudication, keeping verification independent of the code that produces candidates; documented a genuine unresolved miss rather than tuning it away.
- Salesforce NL→SOQL Dashboard** — Natural-Language Analytics | *Python, FastAPI, React, Salesforce SOQL, Claude API*
- Built a schema-aware natural-language-to-SOQL agent that converts plain-English questions into validated Salesforce queries for non-technical users.
 - Validated generated queries against live org metadata, catching references to objects and fields that do not exist before anything reaches the API.
 - Added a validation layer that blocks malformed or unsafe queries **before** execution and feeds the parse error back for a bounded retry, so a bad first attempt self-corrects.
 - Rendered results as live dashboard widgets (tables and charts) with sub-10-second response, turning analyst-written SOQL into self-serve access for business teams.
- RFP Compliance Agent** — Tender Obligation Extraction | *Python, FastAPI, pgvector, MCP, Claude API*
- Extracts binding obligations from 96-page government tenders, separating mandatory clauses (*shall, must*) from discretionary language (*may, desirable*) that carries no compliance weight.
 - Resolves tier-conditional thresholds split across sections, where a requirement’s text and its numeric constraint live in separate tables joined on bidder tier.
 - Routes numeric constraints (turnover, headcount, project value) to deterministic checks and sends only qualitative clauses to the model, keeping model invocations proportional to genuine ambiguity rather than document length.
 - Treats tender PDFs as untrusted input against prompt injection, and reports **false-compliance rate** as the headline metric, since a wrongly-cleared gap disqualifies an entire bid.

ACHIEVEMENTS & ACTIVITIES

- Solved 350+ Data Structures and Algorithms problems across LeetCode and other coding platforms.
- Represented the school at the State-level Basketball Championship.